Implement a CSS grid-based layout

Aim:

To design and implement a web page layout using the CSS Grid system in HTML and CSS to organize webpage elements in rows and columns.

Description:

This program demonstrates how to create a structured webpage layout using CSS Grid. CSS Grid is a powerful layout system that allows developers to arrange web elements into rows and columns easily. In this program, different sections of a webpage such as the header, navigation bar, content area, sidebar, and footer are arranged using the grid layout. CSS properties like `display: grid`, `grid-template-areas`, `grid-template-columns`, and `grid-template-rows` are used to control the placement and size of each section. This method helps in creating clean, flexible, and responsive layouts that adjust well across different screen sizes. The program illustrates how CSS Grid simplifies complex webpage layouts and improves the overall structure of web design.

Algorithm:

1. Start the program.
2. Create the basic HTML structure using `<html>`, `<head>`, and `<body>` tags.
3. Add a title for the webpage in the `<title>` tag.
4. Create a main container `<div>` that will act as the grid container.
5. Inside the container, add different sections such as header, navigation, main content, sidebar, and footer.
6. Apply CSS to the container and set `display: grid` to enable the grid layout.
7. Define the grid structure using `grid-template-columns` and `grid-template-rows`.
8. Use `grid-template-areas` to assign specific positions for each webpage section.
9. Apply styles such as background colors, padding, and margins to make each section visually distinct.
10. Ensure proper alignment and spacing between grid elements.
11. Save the HTML file and open it in a web browser.
12. Verify that the webpage layout is arranged correctly using the CSS Grid system.

13. Stop the program.

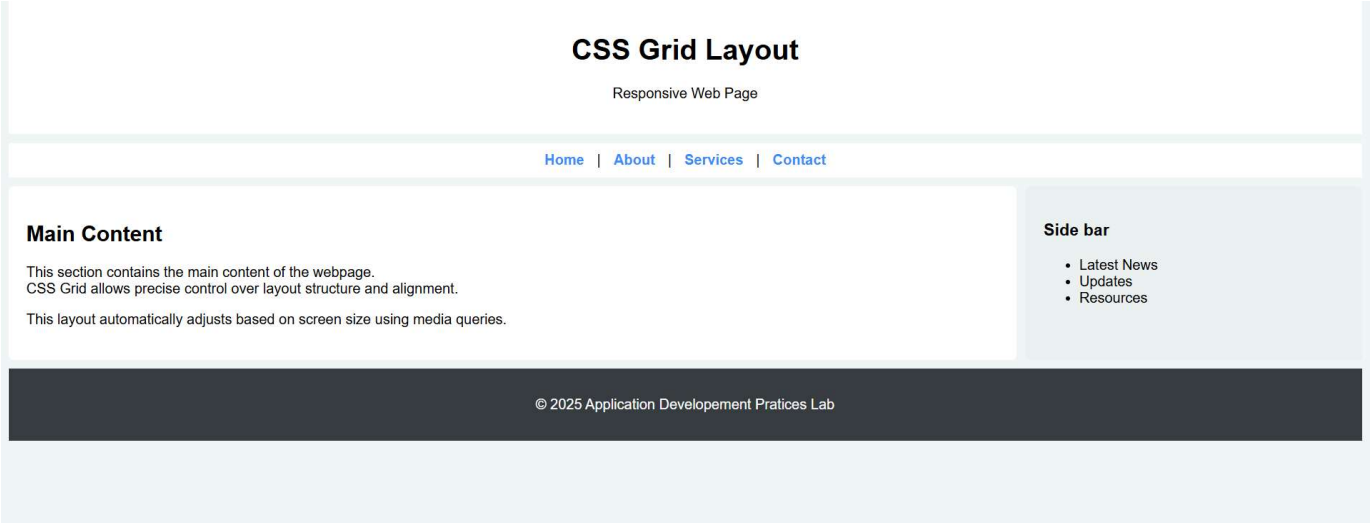
Code:

```
<!DOCTYPE html>
<html>
    <head>
        <title>
            Experiment 3
        </title>
        <style>
        body {
            margin : 0;
            font-family: Arial, Helvetica, sans-serif;
            background-color: #f4f6f8;
        }
        .grid-container{
            display: grid;
            grid-template-areas:
            "header header"
            "nav nav"
            "content sidebar"
            "footer footer";
            grid-template-columns: 3fr 1fr;
            grid-template-rows: auto;
            gap: 10px;
            padding: 10px;
        }
        header{
    grid-area:header;
    background-color:#ffffff;
    color:rgb(0, 0, 0);
    padding:20px;
    text-align:center;
}
nav{
    grid-area:nav;
    background-color:#ffffff;
    padding:10px;
```

```
    text-align:center;
}
nav a{
    color:rgb(61, 141, 254);
    text-decoration:none;
    margin:0 10px;
    font-weight:bold;
}
nav a:hover{
    text-decoration:underline;
}
main{
    grid-area:content;
    background-color:white;
    padding:20px;
    border-radius:5px;
}
aside{
    grid-area:sidebar;
    background-color:#ecf0f1;
    padding:20px;
    border-radius:5px;
}
footer{
    grid-area:footer;
    background-color:#393d42;
    color:white;
    text-align:center;
    padding:15px;
}
            </style>
    </head>
    <body>
        <div class="grid-container">
        <header>
            <h1>
                CSS Grid Layout
            </h1>
            <p>Responsive Web Page</p>
        </header>
        <nav>
```

```html
            <a href="#">Home</a> |
            <a href="#">About</a> |
            <a href="#">Services</a> |
            <a href="#">Contact</a>
        </nav>
        <main>
            <h2>Main Content</h2>
            <p>
                This section contains the main content of the webpage.<br>
                CSS Grid allows precise control over layout structure and
alignment.
            </p>
            <p>
                This layout automatically adjusts based on screen size using
media queries.
            </p>
        </main>
        <aside>
            <h3>Side bar</h3>
            <ul>
                <li>Latest News</li>
                <li>Updates</li>
                <li>Resources</li>
            </ul>
        </aside>
        <footer>
            <p>&copy; 2025 Application Development Practices Lab</p>
        </footer>
        </div>
    </body>
</html>
```

Output :



Result:

The above program has been successfully executed and the output has been verified.